

A proof of the conjectured recurrence for OEIS A157125

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Abstract

OEIS A157125 carries a conjectured linear recurrence with polynomial coefficients, of order 5, contributed by R. J. Mathar. We prove it. The generating function of the entry is algebraic, and the recurrence is equivalent to the vanishing of an explicit differential operator applied to it: writing $\theta = x \frac{d}{dx}$, the sequence $b(n) = \sum_i p_i(n) a(n-i)$ has generating function $B(x) = \sum_i x^i (p_i(\theta + i)A)(x)$, so the conjecture holds for all $n > d$ exactly when B is a polynomial of degree d . Since A lies in a multiquadratic extension of $\mathbb{Q}(x)$, which is closed under θ , the residual B can be computed in closed form. Here it equals $-2x^3 - 6$, a polynomial of degree 3, which proves the recurrence for every $n > 3$.

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1 The sequence and the conjecture

OEIS A157125 is “A transform of the Catalan numbers”. It has offset 0 and begins

$$a(0), \dots, a(7) = 1, -1, -1, 0, 2, 1, -1, -4, \dots$$

The entry gives the generating function

$$A(x) = \sum_{n \geq 0} a(n) x^n = \frac{(x-1) \left(\sqrt{x^2+1} - \sqrt{4x^3+x^2+1} \right)}{2x^3 \sqrt{x^2+1}}, \quad (1)$$

and it carries the following comment:

Conjecture: $(n+3)^*(n-2)^*a(n) - 4^*a(n-1) + 2^*(n^2-n-4)^*a(n-2) + 2^*(2^*n^2-7^*n+2)^*a(n-3) + (n+1)^*(n-4)^*a(n-4) + 2^*(n-1)^*(2^*n-7)^*a(n-5) = 0$. - R. J. Mathar, Nov 15 2012

As of the “Last modified” line on the live entry (2024-10-05, revision 11) the statement is still recorded as a conjecture, and no proof appears among the entry’s links, formulas or programs.

Writing the conjectured relation in the form $\sum_{i=0}^5 p_i(n) a(n-i) = 0$, the coefficient polynomials are

$$p_0(n) = (n-2)(n+3), \quad p_1(n) = -4, \quad p_2(n) = 2(n^2-n-4), \quad p_3(n) = 2(2n^2-7n+2), \quad p_4(n)$$

2 Recurrences as differential operators

Let $A(x) = \sum_{n \geq 0} a(n)x^n$ be the generating function of a sequence, and let $\theta = x \frac{d}{dx}$, so that θ acts on x^m as multiplication by m . For a polynomial p we write $p(\theta)$ for the corresponding differential operator, so that $p(\theta)A = \sum_m p(m)a(m)x^m$.

Lemma 1. Let $p_0, \dots, p_r$ be polynomials and put $b(n) = \sum_{i=0}^r p_i(n) a(n-i)$, with the convention $a(m) = 0$ for $m < 0$. Then

$$B(x) := \sum_{n \geq 0} b(n) x^n = \sum_{i=0}^r x^i (p_i(\theta + i)A)(x).$$

Proof. Fix i and substitute $m = n - i$:

$$\sum_{n \geq 0} p_i(n) a(n-i) x^n = \sum_{m \geq 0} p_i(m+i) a(m) x^{m+i} = x^i \sum_{m \geq 0} p_i(m+i) a(m) x^m = x^i (p_i(\theta + i)A)(x),$$

the last step because $p_i(\theta + i)$ multiplies the coefficient of x^m by $p_i(m+i)$. Summing over i gives the claim. $\square$

Corollary 2. With the notation of Lemma 1, the recurrence $\sum_i p_i(n) a(n-i) = 0$ holds for every $n > d$ if and only if B is a polynomial of degree at most d .

Proof. $b(n)$ is the coefficient of x^n in B , so $b(n) = 0$ for all $n > d$ says exactly that B has no terms above x^d . $\square$

This turns the conjecture into a closed-form identity. The point is that it can then be decided exactly, with no truncation: A is algebraic, so B lies in the same field as A , and one only has to recognise it.

Lemma 3. Let $D_1, \dots, D_k \in \mathbb{Q}(x)$ and let $K = \mathbb{Q}(x)[\sqrt{D_1}, \dots, \sqrt{D_k}]$. For a subset $S \subseteq \{1, \dots, k\}$ write $e_S = \prod_{i \in S} \sqrt{D_i}$, so that the e_S span K over $\mathbb{Q}(x)$. Then K is closed under θ , and θ is diagonal in this spanning set:

$$\theta(c e_S) = \left(x c' + c \sum_{i \in S} \frac{x D_i'}{2 D_i} \right) e_S, \quad c \in \mathbb{Q}(x).$$

Proof. For each i , $(\sqrt{D_i})' = D_i'/(2\sqrt{D_i}) = (D_i'/(2D_i))\sqrt{D_i}$. Applying the Leibniz rule to the product $e_S = \prod_{i \in S} \sqrt{D_i}$ therefore returns e_S itself multiplied by $\sum_{i \in S} D_i'/(2D_i)$, a rational function. Differentiating $c e_S$ and multiplying by x gives the stated formula. No basis element other than e_S occurs, which is the diagonality claim. $\square$

Consequently, if $A \in K$ then every $p_i(\theta + i)A$ is in K , and so is B . Writing $B = \sum_S c_S e_S$ with $c_S \in \mathbb{Q}(x)$, the test of Corollary 2 becomes: B is a polynomial precisely when $c_S = 0$ for every $S \neq \emptyset$ and $c_\emptyset$ has constant denominator. Both are decided by exact cancellation of rational functions.

3 The computation for A157125

The generating function (1) lies in $K = \mathbb{Q}(x)[\sqrt{D_1}, \dots, \sqrt{D_2}]$ with

$$D_1 = x^2 + 1, \quad D_2 = 4x^3 + x^2 + 1.$$

Applying Lemma 1 to the coefficient polynomials of Section 1 and reducing in K by Lemma 3, every component c_S of B with $S \neq \emptyset$ cancels identically and the remaining rational part collapses to

$$B(x) = -2x^3 - 6,$$

a polynomial of degree 3.

